

Poster Presentation Rubric

EEB 187: Ecology and Evolution of Color | Spring 2026

Group:	Clade:
Date:	Grader:

Presentations: Tue Jun 3 (Group A) / Thu Jun 5 (Group B)

Scoring: 8 categories, each on a 4-point scale. Total: **32 points**.

- **4 = Excellent** – exceeds expectations
- **3 = Good** – meets expectations, minor areas to strengthen
- **2 = Adequate** – meets minimum requirements but with notable gaps
- **1 = Developing** – still building toward expectations

Part A: Poster Design and Physical Elements (50%, 16 points)

Category 1: Visual Clarity and Layout

Score	Descriptor
4	Poster follows the 3-column template layout cleanly. Title is readable from 6 feet. Section headers and body text follow a clear font hierarchy. Figures are large and well-positioned. White space is used effectively. The eye flows naturally from Introduction through Results to Conclusions.
3	Layout is organized and mostly readable. The overall flow is clear, but some elements could be improved – a slightly crowded section, a caption that’s hard to read, or a figure that could be repositioned.
2	Layout has noticeable problems that affect readability: text is too small to read at conversational distance, figures are crammed together, or the reading path is unclear.
1	The poster layout is still developing and would benefit from reorganization to improve readability and visual flow (e.g., sections are out of order or the viewer cannot identify the main result at a glance).

Category 2: Figure Quality

Score	Descriptor
4	All figures are poster-ready: high resolution, labeled axes with variable names and units, appropriate color palettes, no default R styling, and informative captions that explain what each figure shows.

Score	Descriptor
3	Figures are clear and high resolution. Most elements are well-executed, but some figures could be improved – a missing axis unit, a brief caption, or one figure with default styling.
2	Figures have noticeable problems that affect interpretation: low resolution, unlabeled axes, default R gray backgrounds, or missing captions.
1	Figures are still in draft form and would benefit from additional polish to meet poster presentation standards (e.g., placeholder figures or screenshots rather than exported graphics).

Category 3: Scientific Content and Narrative

Score	Descriptor
4	The poster tells a coherent scientific story. It includes a clear research question, a hypothesis with testable predictions that connect to the methods used, and an interpretation of whether the results support or reject those predictions. These elements may be combined (e.g., question and hypothesis in a single section) as long as they are all clearly present. The narrative flows logically from question through methods to conclusion.
3	The research question, hypothesis, and predictions are present and the narrative is coherent, but could be sharpened – for example, the prediction is implied rather than stated explicitly, or the connection between the hypothesis and the chosen method is not made clear. The group acknowledges at least one limitation of their approach.
2	The poster has a research question but the hypothesis or predictions are vague or missing. The reader has difficulty seeing how the methods chosen connect to the question being asked.
1	The scientific framing is still developing and would benefit from a clearer statement of what question is being asked, what the group expected to find, and how their methods address that expectation (e.g., methods are described but not connected to a specific hypothesis).

Category 4: Analytical Rigor

Score	Descriptor
4	The group ran at least three analytical recipes, including at least one descriptive analysis (e.g., color gamut, morphospace, discrete trait mapping) and at least one phylogenetic test (e.g., phylogenetic signal, PGLS, model comparison, DTT). Statistical results are accurately reported with appropriate notation. Interpretations are valid – the group distinguishes significant from non-significant results and does not overstate conclusions. Non-significant results are reported honestly and interpreted constructively.
3	Analyses were run and results are mostly reported accurately. The group demonstrates understanding of their statistical output and acknowledges at least one limitation, but some elements could be improved – a missing test statistic or an interpretation that slightly overstates the conclusion.
2	Analyses were run but results are reported incompletely or with errors that suggest uncertainty about what the output means.

Score	Descriptor
1	The analytical component is still developing and would benefit from additional work on running, reporting, and interpreting the chosen analyses (e.g., analyses are described but results are not shown, or numerical results are present without interpretation).

Part B: Oral Presentation (50%, 16 points)

These categories assess whether presenters can explain the reasoning behind their project. Each presenter should be prepared to address any of these topics.

Category 5: Explain Your Primary Descriptive Figure

Score	Descriptor
4	The presenter clearly explains what their main descriptive figure shows – what data went into it, what the axes or categories represent, and what biological pattern it reveals. They explain why they chose this visualization and connect it to an underlying biological concept from the course.
3	The presenter explains most elements of the figure accurately and can describe the biological pattern, but struggles to connect it to broader course concepts or to articulate why this visualization was chosen.
2	The presenter has difficulty explaining what the figure shows. They can identify it by name but cannot describe the axes, categories, or biological meaning without reading from the poster.
1	The presenter is still developing their understanding of the figure and needs significant support to explain what it shows (e.g., reads the caption aloud rather than explaining in their own words).

Category 6: Explain How Your Results Connect to Evolutionary History

Score	Descriptor
4	The presenter explains how their analyses incorporate the phylogeny and why that matters. They can interpret their phylogenetic results in plain English – what the values mean biologically, where on the tree the interesting patterns occur, and what this tells us about how color has evolved in their clade.
3	The presenter explains most of their phylogenetic results accurately and can describe the biological meaning, but struggles to articulate why the phylogeny is necessary or to identify specific evolutionary patterns on the tree. They acknowledge at least one limitation of their phylogenetic approach.
2	The presenter has difficulty explaining their phylogenetic results beyond stating numerical values. They can name the methods used but cannot interpret what the output means for their clade's color evolution.
1	The presenter is still developing their understanding of how the phylogeny connects to their analysis and needs significant support to explain the evolutionary component (e.g., can state that “we used Blomberg’s K” but not what K measures).

Category 7: Motivate and Interpret the Statistical Test

Score	Descriptor
4	The presenter explains why their chosen statistical test requires phylogenetic correction (species are not independent because they share evolutionary history). They state their hypothesis clearly, identify the predictor and response variables, and interpret the result biologically – not just whether it was significant, but what it means for color evolution in their clade. They can explain what it would mean if the result had gone the other way.
3	The presenter explains their statistical test and interprets the result accurately, but struggles to articulate why phylogenetic correction was needed or what the result would mean if it had gone differently.
2	The presenter has difficulty moving beyond reporting p-values. They can state whether the result was significant but cannot explain the biological meaning or why ordinary regression would be inappropriate.
1	The presenter is still developing their understanding of the statistical approach and needs significant support to explain why the test was used or what the result means (e.g., states “the p-value was 0.04” without further interpretation).

Category 8: Delivery, Questions, and Connections

Score	Descriptor
4	The presenter speaks conversationally, makes eye contact, and uses the poster as a visual aid – pointing to figures while explaining rather than reading text. They handle follow-up questions confidently, connecting their clade’s results to course concepts from multiple weeks. They can compare their findings to other clades or published studies and acknowledge limitations honestly. Their answers show genuine understanding beyond what is written on the poster.
3	The presenter communicates clearly and answers most questions accurately, making at least one connection to course concepts beyond their poster. They occasionally read from the poster or need a moment to formulate an answer but demonstrate solid understanding. They acknowledge at least one limitation when asked.
2	The presenter has difficulty answering questions that go beyond what is written on the poster. They tend to read sections aloud rather than explaining conversationally. They can handle factual questions about their own analysis but struggle with “why” or “how does this relate to...” questions.
1	The presenter is still developing their presentation skills and would benefit from practice explaining their project conversationally rather than reading from the poster. They need significant support to connect their project to broader course themes.

Quick Score Sheet

Circle one score per category. Sum for total.

Group:	Clade:
Date:	Grader:

PART A: POSTER DESIGN (16 pts)

#	Category	4	3	2	1	Score
1	Visual clarity and layout	4	3	2	1	_____
2	Figure quality	4	3	2	1	_____
3	Scientific content and narrative	4	3	2	1	_____
4	Analytical rigor	4	3	2	1	_____
Subtotal A:						_____ / 16

PART B: ORAL PRESENTATION (16 pts)

#	Category	4	3	2	1	Score
5	Explain primary descriptive figure	4	3	2	1	_____
6	Connect results to evolutionary history	4	3	2	1	_____
7	Motivate and interpret statistical test	4	3	2	1	_____
8	Delivery, questions, and connections	4	3	2	1	_____
Subtotal B:						_____ / 16

TOTAL: _____ / 32	Percentage: _____ %
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Comments:
